Linear Regression by Optimisation

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Introduction

Linear models act to describe the relationship between a response variable Y and its predictor X. It acts as the best linear approximation to the true relationship.

The true model is represented by the equation:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon$$

where:

- Y_i – i-th response variable
- X_i – i-th predictor variable
- β_0 – true intercept term
- β_1 – true predictor coefficient
- ϵ – random error

Using collected data, we can make the best point estimates for the true value of coefficients and have:

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i + e_i$$

where:

- $\hat{\beta}_0$ – estimated intercept term
- $\hat{\beta}_1$ – estimated predictor coefficient
- e_i – residual

Using Gradient Descent

In order to find the best approximations $\hat{\beta}_0$ & $\hat{\beta}_1$, we can use gradient descent and minimise mean aggregate errors known as mean squared error (MSE). MSE is given by the formula:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

We can utilise partial derivatives to find estimates of β_0 & β_1.

$$f(\hat{\beta}_0, \hat{\beta}_1) = \frac{1}{n}\sum_{i=1}^{n}\left[Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i + e_i)\right]^2$$

$$\frac{\partial f}{\partial \hat{\beta}_0} = \frac{2}{n}\left[Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i + e_i)\right] \qquad \frac{\partial f}{\partial \hat{\beta}_1} = \frac{2}{n}X_i\left[Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i + e_i)\right]$$

Now to reach optimal values of $\hat{\beta}_0$ & $\hat{\beta}_1$ that minimise errors e_i, where $e_i = Y_i - \hat{Y}_i$, we use gradient descent iteratively by taking steps L in the negative direction of max gradient.

$$\hat{\beta}_{0(\text{new})} = \hat{\beta}_0 + L\frac{\partial f}{\partial \hat{\beta}_0} \qquad \hat{\beta}_{1(\text{new})} = \hat{\beta}_1 + L\frac{\partial f}{\partial \hat{\beta}_1}$$